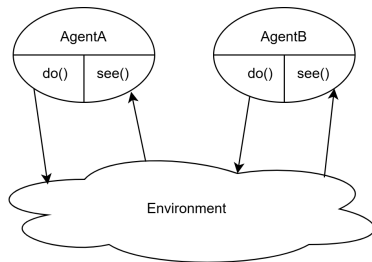


BIOLOGY BACKGROUND

- ▶ Parkinson's disease is characterized by progressive loss of dopaminergic neurons in the substantia nigra.
- ▶ Reduced dopamine output disrupts motor control and produces symptoms such as tremor and rigidity.
- ▶ Misfolded α -synuclein can aggregate into toxic structures and Lewy bodies.
- ▶ Neuroinflammation, mitochondrial dysfunction and impaired degradation mechanisms contribute to disease progression.

STIGMERGIC SYSTEMS

- ▶ **Stigmergy** is an approach that describes indirect coordination through modifications of a shared environment.
- ▶ Agents do not communicate directly: they influence and are influenced by environmental traces.



- ▶ In **biological systems**, interactions are mediated by local signals and fields, allowing systems to capture how small local changes propagate and amplify over time.

ADAPTIVE AGENTS

- ▶ Agents are modeled as **adaptive agents**.
- ▶ Their actions depend on their internal state and perception.
- ▶ No global controller declared, the behavior logic emerges from the local entities.
- ▶ The environment works only as a bridge between the agents.

An adaptive agent is defined as:

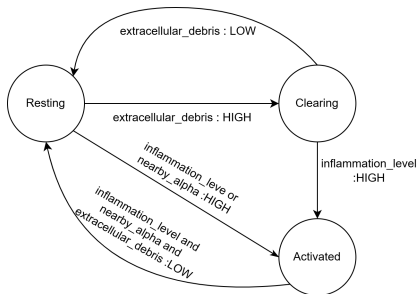
- ▶ Sets:
 - ▶ I , internal states
 - ▶ E , environmental states
 - ▶ Per , perceptions
 - ▶ Ac , possible actions
- ▶ i_0 , initial state
- ▶ Functions:
 - ▶ $see : E \rightarrow Per$
 - ▶ $next : I \times Per \rightarrow I$
 - ▶ $action : I \times Per \rightarrow Ac$
 - ▶ $do : Ac \times E \rightarrow E$

Env (SUBSTANTIA NIGRA)

- ▶ Environment lived by the **extra-cellular** agents (Dopaminergic Neurons, Microglia, Astrocytes, α -synuclein).
- ▶ **Discrete Space:** It's represented as a discrete grid.
- ▶ **Goal:** Highlight the release of dopamine and the transmission of the disease between neurons.
- ▶ **Fields:** extracellular_debris, inflammatory_levels, dopamine_output

Ag_{Mic} (MICROGLIA)

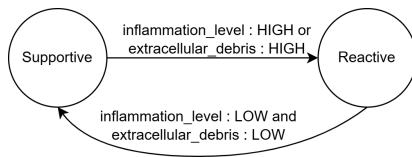
- ▶ **Internal States:** Resting, Clearing, Activated
- ▶ **Action:** Scan, Clear_Debris, Release_Inflammation
- ▶ **Per:** position, extracellular_debris, inflammatory_levels, nearby_alpha



$action(Resting, (-)) = Scan$
 $action(Clearing, (-)) = Clear_Debris$
 $action(Activated, (-)) = Release_Inflammation$

Ag_{Ast} (ASTROCYTE)

- ▶ **Internal States:**
Supportive, Reactive
- ▶ **Action:** Provide_Support, Release_Inflammation
- ▶ **Per:** position, inflammation_level, extracellular_debris



$action(Supportive, (-)) = Provide_Support$

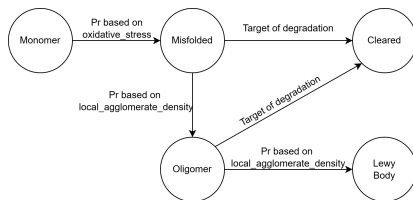
$action(Reactive, (-)) = Release_Inflammation$

Ag_{Neu} (DOPAMINERGIC NEURON) AS ENVIRONMENT

- ▶ **MacroAgent:** Acts as both an agent and an environment, it contains lower level (intracellular) agents through composition.
- ▶ **Complex State:**
 - ▶ list of agents
 - ▶ scalar fields
 - ▶ 10×10 grid in which all the agents move
 - ▶ buffers for lysosome targetting
- ▶ **Goal:** Simulate the evolution of the pathology within the neuron and the degeneration of its health.
- ▶ Perceives both the external environment and its internal one.

Ag_{α} (α -SYNUCLEIN)

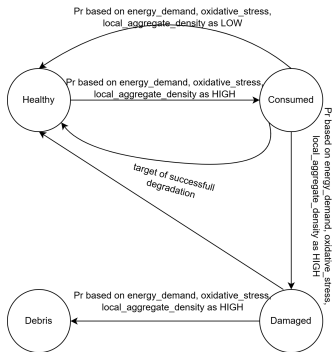
- ▶ **Internal States:** Monomer, Misfolded, Oligomer, Cleared, LewyBody
- ▶ **Action:** Move, Stay
- ▶ **Per:** position, oxidative_stress, neighborhood, local_aggregate_density



$action(Monomer, (-)) = Move \vee Stay$
 $action(Misfolded, (-)) = Move \vee Stay$
 $action(Oligomer, (-)) = Move \vee Stay$
 $action(LewyBody, (-)) = Stay$
 $action(Cleared, (-)) = Stay$

Ag_{mito} (MITOCHONDRION)

- ▶ **Internal States:** Healthy, Consumed, Damaged, Debris
- ▶ **Action:** Stress, Fuse, Divide
- ▶ **Per:** position, oxidative_stress, energy_demand, local_aggregate_density, local_debris_density, target_assigned



$action(Healthy, (energy_demand : HIGH)) = Stress$
 $action(Consumed, (ldd, lad : LOW)) = Fuse$
 $action(Consumed, (-)) = Stress$
 $action(Damage, (-)) = Divide$

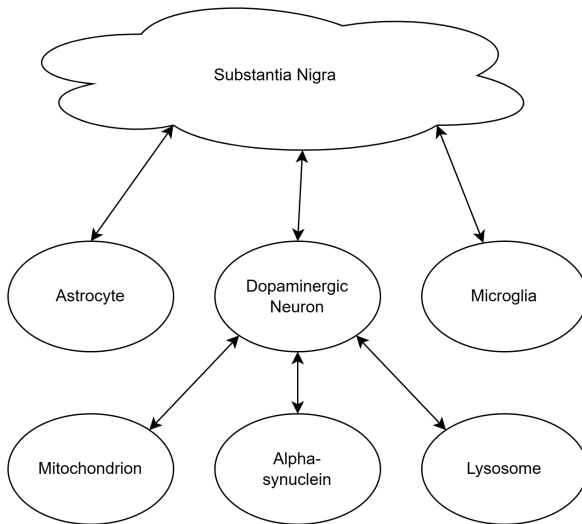
Ag_{lyso} (LYSOSOME)

- ▶ **Internal States:** Inactive, Active, Overwhelmed
- ▶ **Action:** Scan, Select_Target, Degrade
- ▶ **Per:** position, targets, tasks, local_aggregate_density



$action(Inactive, (-)) = Scan$
 $action(Active, (target : empty)) = Select$
 $action(Active, (task : empty)) = Degrade$

FULL SIMULATION ARCHITECTURE



GOAL OF THE MULTI-LEVEL ANALYSIS

Main idea

Start from a complete timed causal trace and progressively contract the graph to reveal higher-level mechanisms and recurring structural patterns.

$$G_0 \rightarrow G_1 \rightarrow G_2 \rightarrow G_3$$

- ▶ G_0 : full timed causal trace.
- ▶ G_1 : time-contracted graph.
- ▶ G_2 : agent-clustered graph.
- ▶ G_3 : structural motif graph.

OVERVIEW OF THE FOUR GRAPH LEVELS

Level	Contraction principle	Interpretation
G_0	None	Full execution trace
G_1	Remove time index	Recurring state interactions
G_2	Merge agent identities	Agent-type mechanisms
G_3	Contract topological motifs	Relations among patterns

Progressive abstraction

Each level preserves selected causal information while removing details that are not needed for the next scale of analysis.

G_0 DEFINITION: TEMPORAL CAUSAL GRAPH

- ▶ **Node definition:** either an agent in a state at a time
(AgentNameID_State@time)
or an environment field at a time
(EnvironmentField@time)
- ▶ **Edge definition:** directed edges encode causal influence.
 - ▶ agent at moment $t \rightarrow$ agent at moment $t + 1$;
 - ▶ agent $\rightarrow$ environment variable;
 - ▶ environment variable $\rightarrow$ agent;

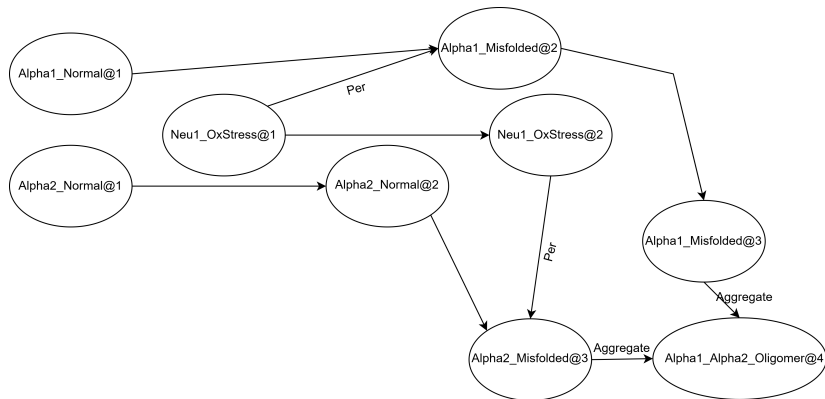
Simplification

For simplification purposes, some mechanisms were directly represented at agent level.

- ▶ some agent transition;
- ▶ lysosome targeting relation;
- ▶ aggregation;

NOTE: G_0 is always acyclic.

G_0 EXAMPLE



G_1 : TIME-CONTRACTED GRAPH

Contraction rule

Collapse nodes that differ only by time:

$$x@1, x@2, x@3 \mapsto x$$

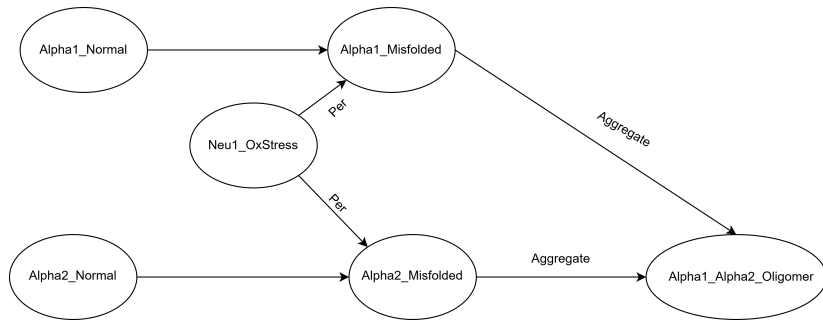
Analytical role

G_1 captures recurring causal relations between states of the same concrete agents and environment fields, independent of when they occur. It can identify state-level feedback loops and recurrent causal dependencies.

Consequence

Removing time can create cycles, because repeated temporal interactions are projected onto the same node set.

G_1 EXAMPLE



G_2 : AGENT-CLUSTERED GRAPH

Contraction rule

Collapse individual agent identities while preserving agent type (A) and state (S):

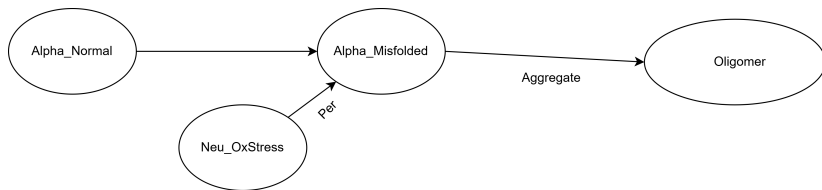
$$A1_S, A2_S \mapsto A_S$$

Analytical role

G_2 reveals how classes of agents interact with each other and with environmental variables, independently of specific agent instances.

- ▶ Removes individual IDs.
- ▶ Aggregates repeated mechanisms across many agents.

G_2 EXAMPLE



G_3 : STRUCTURAL PATTERN GRAPH

Contraction rule

Contract relevant topological structures in G_2 , such as:

- ▶ cycles;
- ▶ cliques;
- ▶ strongly connected components;
- ▶ repeated causal paths;

Interpretation

G_3 is a graph of patterns. Nodes no longer represent individual states, but higher-order causal structures.